



PROFILE

I am a motivated engineering student with strong interests in machine learning, energy optimization, and software development, and hands on experience building data driven solutions. My work includes deep learning based solar panel fault detection, ML-driven energy optimization for Electric Arc Furnaces, and software development using Java and web technologies for real world applications.

I am seeking a technical internship where I can apply my skills in machine learning, data analysis, and software engineering to scalable, production oriented systems.

TECH SKILLS

- Java
- C
- Python
- MATLAB
- JavaScript
- HTML
- CSS

CONTACT

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<https://www.linkedin.com/in/ranjith-raja-2405a7343/>

Github:

<https://github.com/Ranjith-7777>

Portfolio:

<https://portfolio-9uoygl477-ranjith-7777s-projects.vercel.app/#home>

RANJITH RAJA B

ENGINEERING UNDERGRADUATE [2nd Year]



EDUCATION

- **Bachelor's Degree in Computer Science and AI** 2024-Present
School of AI | Amrita Vishwa Vidyapeetham
CGPA: 9.41/10
- **School Education** 2011-2024
The Vikasa School, Tuticorin
 - **Class 12(ISC)** – 92.2%
 - **Class 10(ICSE)** – 97.2%



PROJECTS

- **Campus Navigation System** [COMPLETED]
Technologies Used: HTML, CSS, JavaScript, Spring Boot.
Description: Developed an interactive campus navigation system with real time location tracking and an optimised user interface. Spring boot for backend support.
- **Solar Panel Fault Detection using TripleNet** [COMPLETED]
Technologies Used: Python, tensorflow, MATLAB, DeepLearning
Description: Implemented a lightweight deep learning model (TripleNet) to detect faults in solar panels using image processing.
- **Burglar Detection System** [COMPLETED]
Technologies Used: Arduino Uno, RFID, Bluetooth module (HC-05), Arduino IDE
Description: Developed a security system using RFID technology combined with the Bluetooth communication to detect the invader.
- **Optimization of Energy in Electric Arc Furnace** [COMPLETED]
Researched ways to optimize energy consumption in EAF. Applying machine learning techniques to improve efficiency based on process data.
- **Operating System (Mini-OS)** [COMPLETED]
Designed and implemented a basic operating system from scratch as part of systems coursework.



RESEARCH EXPERIENCE

- Published three peer-reviewed research papers in ESPR, Elsevier: Energy Conservation and Management, and IEEE Access.
- Proposed TripleNet, a deep metric learning framework for solar photovoltaic fault detection, improving robustness under imbalanced and noisy conditions.
- Developed a GRU Pyramid architecture to capture multi resolution temporal patterns for solar panel fault classification, achieving improved detection accuracy and generalisation.
- Conducted a comprehensive ML based energy efficiency analysis of EAFs.



CONFERENCES AND HACKATHONS

- Presented research at IEEE CICC-2025 (NIT Goa) on ML based attack detection in RPL using LightBGM and Dempster-Shafer theory.
- Participated in Smart India Hackathon (SIH) and HackTide, working on AI/software solutions in fast paced, collaborative hackathon environments.